

Detecting departure and destination aerodromes from ADS-B

Version 5.1 — why arrivals are hard, and what recovers part of them

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Summary

Version 4 established that arrivals are harder than departures on every measure and recommended not changing the arrivals algorithm. It did not say *why*, and “we do not know why” is a poor basis for leaving a quarter of the data unlabelled. This version takes the arrival deficit apart.

Three findings.

The arrival deficit is silence, not error. Of 87,152 in-area arrivals, 3,011 are answered wrongly and **21,009 are not answered at all**. Version 4’s emphasis on misnaming — built on a vivid case, Bilbao — described 12.54% of the problem.

Most of that silence is ours, but it is not cheap to break. 13,300 flights had a usable candidate that the abstention refused; only 7,709 had nothing at all. So the recoverable share is nearly twice the permanently lost one. But guessing on the refused flights scores 28.27% — four right for eleven wrong. The abstention is doing its job.

One new signal beats that, and it is horizontal. Whether the aircraft’s course *points at* the candidate aerodrome, measured tightly enough, recovers 425 correct arrivals for only 97 new errors — **4.38:1**, comfortably past the 2:1 exchange rate version 4.5 argued for. This is the first change in the study to clear that bar. It applies to arrivals only: on departures the same test loses at 0.09:1. Vertical rate recovers nothing, and adding it to the bearing test destroys most of the gain.

There is a single reason for that asymmetry, and it is the most useful thing in this paper:

! Positions survive staleness; rates do not

The OpenSky archive fills every second by carrying the last known position forward, and in this population roughly half the rows are repeats. **A bearing between two genuinely measured fixes is correct regardless of when they were measured. A rate divides by an elapsed time the timestamps get wrong** — the clock says ten minutes, the two real measurements may be ninety seconds apart.

That one sentence explains why bearing alignment works, and why vertical rate, closing rate and the approach cone of version 4.5 all failed.

1 Where the arrival deficit sits

Table 1: Arrivals not correctly labelled, by class of the true aerodrome. 31,829 of 95,116 flights over 2025-06-05 to 07.

Class	Flights	Coverage	Accuracy	Correct	Missing	Share of deficit
large airport	80,683	76.11%	96.27%	59,120	21,563	67.75%
out of area	7,964	5.12%	37.99%	155	7,809	24.53%
medium airport	5,983	74.88%	89.55%	4,012	1,971	6.19%
small airport	352	57.67%	0.00%	0	352	1.11%
heliport	134	36.57%	0.00%	0	134	0.42%

Two thirds of the deficit is at large airports and a quarter is out-of-area. This paper addresses the first; the second remains the largest untouched block and is left for a successor.

2 Silence, not error

“Not labelled” hides two situations that need opposite responses: **no candidate existed**, which is a receiver-coverage fact, and **a candidate existed and the rule refused it**, which is a choice. **nearest** never abstains, so its coverage is exactly the rate at which a candidate existed — no new computation is needed to separate them.

Table 2: The 87,152 in-area arrivals, decomposed. In-area means the true destination is a real aerodrome, so out-of-area flights are excluded.

Outcome	Flights	Share
correctly labelled	63,132	72.44%
answered, wrong	3,011	3.45%
candidate existed, abstained	13,300	15.26%
no candidate anywhere	7,709	8.85%

8.85% of in-area arrivals have no candidate at all. That is the hard ceiling: no endpoint-based method can exceed 91.15% coverage on this data, whatever rule it uses.

3 The abstained block is hard, not neglected

The 13,300 refused flights look like free coverage. They are not.

endpoint answers a subset of what **nearest** answers, and on shared flights both rank candidates by the same effective distance, so they return identical answers there. **nearest**’s *extra* correct answers are therefore exactly its score on the refused block: **3,760 of 13,300, or 28.27%**.

Answering them all would add 3,760 correct labels and 9,540 wrong ones — 0.39:1. Against the exchange rate version 4.5 argued for, that is not close. **“Relax the thresholds for arrivals” is now priced, and the price is bad.**

Why are they hard? Because of what an arrival endpoint is. The state-vector decimation study measured that an arrival’s last fix carries a position **measured a median of 190 s earlier** — at approach speed, roughly eight nautical miles back up the approach. The endpoint is not where the flight ended. Position alone cannot identify an aerodrome from eight miles out on a busy terminal area, and the abstention correctly declines to try.

4 Bearing alignment

If position is exhausted, motion might not be. The measure is the angle between **the direction the aircraft actually travelled** over a window and **the direction of the candidate aerodrome**, seen from the far end of that window. Zero degrees means the track runs straight at it.

This involves no extrapolated position — it is an angle between two observed bearings — so the documented failures of trajectory extrapolation in version 2 and the 2025 PRC report do not bind it.

i Why departures need no 180-degree correction

A departing aircraft flies *away* from its aerodrome, so it looks as though the measure must be inverted for departures. It must not, because the origin of the angle already flips with the role.

The window is measured from the arrival’s **last** fix and the departure’s **first**, so its far end is earlier in time for an arrival and later for a departure. The bearing from that far end toward the near end is therefore the course in one case and the reverse course in the other.

The question becomes “*is the aerodrome in the direction I am heading*” for an arrival and “*is the aerodrome in the direction I came from*” for a departure, and both read zero for a correct match.

The distinction matters because the two vertical and horizontal *rates* in the same analysis do carry an explicit per-role sign — they are signed along the time axis, and an angle between two bearings from a common origin is not. Confirmed empirically: correctly identified departures sit at a median of 2–9 degrees, not 171–178.

It also explains part of why the measure is weak for departures. By 30 minutes down-track the separation is 6 degrees against 2 — from far enough away, the true origin aerodrome and its neighbours all lie in much the same direction, so the angle stops discriminating. An arrival endpoint sits a median of eight nautical miles short of the runway, where the direction along the track is still informative.

Table 3: Correct labels per error on the abstained arrival block, by window and angle gate. The block’s own baseline is 0.37. Note that these are computed on in-area truth only, and so overstate the population-level trade – see the next section.

Window	1°	2°	3°	5°	10°	20°	30°
2 min	0.44	0.38	0.53	0.41	0.35	0.33	0.29
3 min	0.60	0.49	0.58	0.46	0.40	0.36	0.31
5 min	2.81	2.31	2.10	1.78	1.33	1.13	0.92
7 min	3.14	2.66	2.52	2.38	1.75	1.21	0.96
10 min	1.99	2.45	2.41	1.92	1.40	0.99	0.78
15 min	2.09	1.97	1.79	1.52	1.07	0.69	0.55
20 min	2.47	1.85	1.61	1.27	0.85	0.56	0.49
30 min	2.25	1.60	1.30	0.96	0.61	0.48	0.43

Three things are visible and all three make physical sense.

Short windows carry no information. At 2–3 minutes every cell sits near the 0.37 baseline. Over two minutes the aircraft’s displacement is small against its position error, so the bearing is noise.

It switches on at 5 minutes and peaks at 7. A metric that were an artefact would not have an interior optimum with noise on both sides.

The gate should tighten as the window lengthens. At loose gates short windows win, because over half an hour everything points roughly everywhere. At 1° that reverses — a long baseline gives a more precise bearing, so a tight gate becomes meaningful rather than noise.

Accuracy plateaus at 70–76% however tight the gate. Roughly one arrival in four that points straight at an aerodrome lands somewhere else. Bearing alone has a ceiling, and tuning it further is close to exhausted.

5 The comparison that matters

The sweep above scores the abstained block with out-of-area truth excluded. That flatters it: rescuing a flight whose true destination is outside the observed area produces a wrong answer the block analysis never counted. The honest test is the whole population.

Table 4: Arrivals, all 95,116 ground-truth flights. *Net @2:1* prices each new error at two correct labels, the exchange rate of version 4.5; positive means the gate is worth adopting on that reasoning.

Model	Coverage	Accuracy	Overall	d correct	d wrong	Ratio	Net @2:1
base (30 NM / 15,000 ft)	69.10%	95.63%	66.08%	—	—	—	—
7 min / 0.10 deg	69.65%	95.52%	66.53%	425	97	4.38:1	231
7 min / 0.25 deg	69.83%	95.44%	66.65%	536	161	3.33:1	214
7 min / 0.50 deg	69.96%	95.33%	66.70%	584	237	2.46:1	110
7 min / 0.75 deg	70.05%	95.27%	66.74%	623	283	2.20:1	57
7 min / 1 deg	70.14%	95.22%	66.79%	676	316	2.14:1	44
7 min / 2 deg	70.42%	95.07%	66.94%	817	434	1.88:1	-51
7 min / 3 deg	70.68%	94.92%	67.09%	957	546	1.75:1	-135
20 min / 1 deg	71.59%	94.52%	67.67%	1,510	862	1.75:1	-214
30 min / 1 deg	73.13%	93.66%	68.50%	2,295	1,537	1.49:1	-779
7 min / 1 deg +rate	69.39%	95.39%	66.19%	102	171	0.60:1	-240
7 min / 3 deg +rate	69.60%	95.22%	66.27%	182	293	0.62:1	-404

Table 5: Departures, the same models. Every gate is a loss.

Model	Coverage	Accuracy	Overall	d correct	d wrong	Ratio	Net @2:1
base (30 NM / 15,000 ft)	80.96%	98.16%	79.47%	—	—	—	—
7 min / 0.10 deg	81.06%	98.05%	79.48%	8	89	0.09:1	-170
7 min / 0.25 deg	81.12%	98.00%	79.50%	29	130	0.22:1	-231
7 min / 0.50 deg	81.23%	97.93%	79.55%	79	180	0.44:1	-281
7 min / 0.75 deg	81.34%	97.86%	79.60%	127	242	0.52:1	-357
7 min / 1 deg	81.47%	97.77%	79.65%	173	312	0.55:1	-451
7 min / 2 deg	82.06%	97.25%	79.80%	313	733	0.43:1	-1,153
7 min / 3 deg	82.44%	97.01%	79.97%	481	931	0.52:1	-1,381
20 min / 1 deg	82.28%	97.03%	79.83%	347	912	0.38:1	-1,477
30 min / 1 deg	82.58%	96.80%	79.94%	450	1,094	0.41:1	-1,738
7 min / 1 deg +rate	81.29%	97.95%	79.63%	150	171	0.88:1	-192
7 min / 3 deg +rate	81.94%	97.54%	79.92%	432	505	0.86:1	-578

Because the bearing test only ever *adds* answers — a flight that already passed the abstention is untouched, and the aerodrome named is the same candidate either way — every newly answered flight falls in exactly one of the two delta columns. Their sum is the number of flights the gate rescued, and

$$\text{accuracy on the rescued} = \frac{\Delta_{\text{correct}}}{\Delta_{\text{correct}} + \Delta_{\text{wrong}}}$$

is how often the gate is right *when it overrides the abstention*. It is not the model’s accuracy, which is swamped by the flights the change never touched: at 3° the gate is right 63.67% of the time on the 1,503 flights it rescues, while the model’s accuracy reads 94.92%. The benchmark to hold it against is 28.27% — what guessing blind on that population scores (Section 3).

The gate must be tight, and tighter than seems physical. The ratio climbs monotonically as the gate narrows — 1.75:1 at 3°, 2.14:1 at 1°, 4.38:1 at 0.1° — and accuracy on the rescued flights rises with it, from 63.67% to 81.42%. Under 2:1 pricing the best gate is the tightest tested.

i A prediction this refuted

Before running the sub-degree gates we argued they could not help. A runway complex spans 2–4 km, which from 40 NM subtends 2–3°, so a gate below a degree is tighter than the target: it should stop selecting for “heading to this airport” and start selecting for “aimed at the arbitrary reference coordinate OurAirports publishes”. Accuracy was predicted to plateau and counts to collapse into noise.

Accuracy did not plateau. It rose from 63.67% at 3° to 81.42% at 0.1°, monotonically, on 522 flights — not a handful.

The reasoning was wrong about what the gate selects. It is not resolving which runway an aircraft is aimed at; it is separating tracks *deliberately* pointed at a field, over a long straight leg, from tracks that happen to line up by coincidence. Coincidental alignment is common at 3° and rare at 0.1°, and that is the whole of the discrimination.

Volume and rate trade against each other, and the rate wins. The widest gate tested, +bearing 30 min / 1 deg, recovers 2,295 correct labels — five times the tightest — but at 1.49:1 it costs 779 on the 2:1 reckoning. Chasing volume here is what the exchange rate exists to prevent.

Departures lose. The same test adds 8 correct departures for 89 errors, 0.09:1. Consistent with everything this study has found: departures are not short of evidence, and adding evidence to a rule that is not short of it costs accuracy. **Bearing should not be applied to departures.**

⚠ The block-level ratios were optimistic, and this is why

The abstained-block sweep reports 2.52:1 at 7 min / 3°. The same rule on the full population gives 1.75:1 — the same gate, the same rule, a third less. The difference is entirely the out-of-area flights the block analysis filtered out and the population cannot.

The lesson generalises: a rule evaluated on the subset it was designed for will beat its own field performance, and by a wide margin. Every ratio quoted in Section 4 should be read as an upper bound.

6 Why the vertical channel is dead

Vertical rate is the obvious second signal, and it does not work. Two independent measures of the same quantity — the broadcast `vert_rate`, and an ordinary least squares fit of altitude on time across the window — agree in sign at close to chance.

Table 6: Sign agreement between vertical measures on the abstained arrival block. 50% is chance.

Window	vert_rate vs		rows with vert_rate =	
	OLS	2-point vs OLS	0	rows fresher than 5 s
5 min	43.04%	47.13%	44.85%	10.58%
10 min	51.88%	59.80%	45.29%	28.87%
20 min	57.56%	64.28%	46.70%	43.92%

Table 7: The same agreement at 20 minutes, banded by the share of rows carrying a position fresher than 5 s. Agreement tracks freshness, which identifies the cause rather than inferring it.

Window freshness	Flights	vert_rate vs OLS	2-point vs OLS
<30% fresh	11,960	49.76%	55.39%
30-60%	10,451	59.71%	60.43%
60-90%	11,309	62.50%	77.51%
>=90% fresh	1,067	71.79%	61.39%

Agreement rises with freshness, and 44.85% of rows carry a vertical rate of exactly zero — the signature of a value held over rather than measured. **There is no vertical signal to extract here in either form.**

The measured consequence is in the comparison table: adding the vertical test to the bearing test cuts the arrival gain from 425 correct to 102, and the ratio from 4.38:1 to 0.60:1. It does not merely fail to help — **it discards flights the bearing test correctly rescued.**

7 Recommendation

Adopt the bearing test for arrivals, at a tight gate. +bearing 7 min / 0.10 deg recovers 425 correct arrival labels for 97 new errors — 4.38:1, against the 2:1 bar. Arrival coverage rises +0.55 pp and overall correctness +0.45 pp, for -0.11 pp of accuracy.

This is the first change this study has recommended adopting since version 4 moved detection into the pipeline, and the first to clear the exchange rate on its own terms rather than by argument.

Do not apply it to departures. Every gate tested is a loss, the best at 0.55:1. Departures are not short of evidence, and adding evidence to a rule that is not short of it costs accuracy.

Do not pursue vertical rate on this data. Adding it to the bearing test cuts the arrival gain from 676 correct to 102 and the ratio from 2.14:1 to 0.60:1. It discards flights the bearing test correctly rescued.

Validate on a second period before publishing the change. Everything here is one three-day sample, and the gate was chosen from that sample. 91.15% remains the ceiling regardless.

And out-of-area is still the larger prize. It is 24.53% of the arrival deficit against the 1.34% this recovers. The bearing test is worth having; it is not the main event.

8 Bearing on the whole sample

Used as a rescue, the bearing test can only *add* answers. That is a structural limit rather than a property of the data: a flight the abstention already accepts is untouched, so the 2,870 arrivals the base rule gets **wrong** were unreachable by every variant in Section 5. No rescue can raise accuracy.

Applying the test to every flight opens three questions the rescue framing cannot ask.

- **Veto** — require alignment on flights the gate accepts, so bearing can remove answers. The only variant that can raise accuracy.
- **Replace** — alignment instead of the distance and height gate, testing whether that box carries anything the course does not.
- **Rerank** — choose *which* aerodrome by alignment rather than by effective distance. Version 4.5 argued the abstention is a gate and never a naming rule, and that arrivals fail by misnaming; this is the naming rule that argument implies.

Table 8: Arrivals: the bearing test used four ways. Only the rescue helps; the other three are catastrophic.

Model	Coverage	Accuracy	Overall	d correct	d wrong
base	69.10%	95.63%	66.08%	—	—
rescue: gate OR align $\leq$ 0.1	69.66%	95.52%	66.54%	436	97
rescue: gate OR align $\leq$ 3.0	70.75%	94.93%	67.16%	1,020	544
veto: gate AND align $\leq$ 3.0	22.78%	98.17%	22.36%	-41,585	-2,473
veto: gate AND align $\leq$ 0.1	11.67%	97.30%	11.36%	-52,052	-2,570
replace: align $\leq$ 3.0 only	24.42%	95.95%	23.44%	-40,565	-1,929
replace: align $\leq$ 0.1 only	12.23%	96.59%	11.82%	-51,616	-2,473
rerank by alignment, then gate	35.81%	86.50%	30.97%	-33,396	1,729
rerank, gate OR align $\leq$ 0.1	37.42%	84.44%	31.60%	-32,800	2,667

All three fail, and not marginally. The veto costs 41,585 correct arrivals at 3°. Replace costs 40,565. Rerank costs 33,396 correct labels *and* adds 1,729 wrong ones, dropping accuracy from 95.63% to 86.50%.

8.1 Why: the measure is out of domain on flights that landed

The diagnosis is not that alignment is a weak signal. It is that the flights the gate already accepts are the wrong population to ask.

	no course computable	median alignment	within 3°	on ground at the endpoint
gate accepts	23.8%	21°	18.4%	43.1%
gate refuses	24.3%	53°	5.2%	0.6%

43% of accepted arrivals are on the ground at their endpoint. For those, the last seven minutes are the approach, the landing roll and the taxi to stand — and the “course” over that window is dominated by ground manoeuvring, which points wherever the taxiways happen to run. A further quarter have no computable course at all. A veto at 3° therefore rejects roughly four in five flights it should keep.

The bearing test was developed on the abstained block, where every endpoint is airborne and en route by construction. **Its domain is airborne endpoints, and the flights the gate accepts are mostly not that.** Used as a rescue it never leaves its domain, which is why it works there and nowhere else.

8.2 Rerank refutes the naming-rule argument

The reranking result deserves its own note, because version 4.5 predicted the opposite. Its argument was that the abstention decides only *whether* to answer and never *what* to answer, that arrivals fail predominantly by misnaming, and that a trajectory-based naming rule was therefore the most promising fix available.

Ranking candidates by alignment instead of distance is that rule, and it is far worse than ranking by distance. The reason is geometric: **an aerodrome 200 NM directly ahead is perfectly aligned**. Alignment has no sense of scale, so as a ranking criterion it systematically prefers distant aerodromes that happen to lie on the track. Distance was doing enormous work that the argument did not credit.

Alignment is a **filter over distance-ranked candidates**, not a ranking criterion. That is what the rescue variant does, and it is the only form in which the signal is worth anything.

i A small correction to Section 5

The whole-sample run computes alignment without requiring an altitude, because alignment is purely horizontal; Section 5's figures came from a routine that also fitted the vertical slope and so discarded samples with no altitude. The rescue numbers here are consequently a little higher — 436 correct against 425 at the same gate. The direction and the recommendation are unchanged; the earlier figures were conservative.

9 What is now ruled out

Recorded so it is not re-attempted:

- **Relaxing the abstention thresholds** — priced at 0.39:1 (Section 3).
- **The approach cone** — version 4.5 measured it against the rectangle at matched coverage; departures lost up to 2.55 pp of accuracy, arrivals were a wash.
- **Bucket decimation** — the state-vector study measured +55 correct arrivals per day for a change that rewrites `track_id` for every published row.
- **Vertical rate, in any form** (Section 6).
- **Trajectory extrapolation** — version 2 and the 2025 PRC report, independently.
- **Bearing as a veto, a replacement, or a naming rule** (Section 8). Only the rescue form works, and only because it stays inside the measure's domain.

10 Limitations

- **One sample.** Three days, 2025-06-05 to 07. The operating points were chosen as the argmax over 120 grid cells of that same sample, so the ratios are optimistic. The 2° and 3° columns peak cleanly at 7 minutes, which is why the *region* is credible; the specific numbers are not validated.
- **Two measurement bugs were found in this work**, both by sanity checks rather than by anything failing: `angle_between` used %, and Spark's remainder keeps the sign of the dividend, so bearings 20° apart measured 340° whenever the pair straddled north — the tell was a median alignment of 186° on a 0–180° scale. And the altitude slope divided an altitude change by an elapsed time drawn from different rows, because `min_by` skips

nulls where `min` does not. Both are fixed; both had already produced numbers that were reported before being caught.

- **The $\geq 90\%$ fresh band reverses the freshness trend** for the 2-point measure, on about a thousand flights. Either small-sample noise or something real about very short tracks. Not resolved.
- **nearest and endpoint are assumed to answer identically where both answer**, which underpins Section 3. It follows from both ranking on the same effective distance, but it is an argument, not a measurement.

11 Reproducing

```
# the abstained block, its motion profile, and the angle x window sweep
python benchmarks/abstained_vertical.py --results-dir <dir>

# which vertical measure to trust, banded by freshness
python benchmarks/vertical_measure.py --results-dir <dir>

# base against the motion variants, whole population
python benchmarks/motion_model_compare.py --results-dir <dir>
```

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